RECOVERY LIVES IN U, CONDITIONING ACTS IN R: WHERE ARBITRARY-SCALE SUPER-RESOLUTION GAINS COME FROM

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ABSTRACT

Arbitrary-scale super-resolution (ASISR) networks pair an image encoder with an implicit decoder that also receives the target pixel size (cell). Decoder and cell designs have multiplied, yet their gains shrink at large scales. We explain this with two frequency coordinates. Measured per band, recovery is a function of frequency normalised by the low-resolution Nyquist limit, u. It is the same for four decoder designs on a shared encoder (within 0.01). Training and a stronger encoder raise it only for $u \geq 1$. The cell, in contrast, acts on the output as a low-pass in high-resolution frequency r. For large mismatch, its cost is predicted by the recovered energy at the frequencies it attenuates. Because a fixed r moves to larger u as scale grows, where little is recovered, the cell's leverage vanishes. Doubling the cell costs 2.1–2.5 dB at $\times 2$, 0.13 dB at $\times 4$ and at most 0.03 dB at $\times 6$. All tests were pre-registered with frozen predictions; failed predictions are reported. Matched training over two seeds confirms that a stronger encoder (RDN, SwinIR) raises recovery only above the LR Nyquist limit (+0.07 to +0.30 dB) and that removing the cell costs 0.10 dB at $\times 2$ but nothing measurable from $\times 3$ on. Decoder *design* barely matters, whereas cutting decoder *capacity* costs 0.11–0.21 dB at every scale, which contradicts our frozen prediction.

***Index Terms*—** Arbitrary-scale super-resolution, implicit neural representation, frequency analysis, scale conditioning

1. INTRODUCTION

LIIF [1] reconstructs an image at continuous coordinates. It decodes local encoder features together with the query offset and the target cell size. Many decoders followed: LTE's local Fourier estimator [2], integrated positional encoding [3], attention-based decoders [4], neural operators [5], Gaussian splatting [6] and filtered heat fields [7]. A recent benchmark reports saturation, with gains predictable from model complexity, compute and data [8]. A spectral decoder with a learned global bandwidth also reports leads that shrink at large scales without explaining them [9]. Frequency analyses describe low-pass behaviour [10], probe responses [11] or scale anchoring [12]. They do not say *where* a decoder or the cell can still change the output, or why that space disappears as the scale grows.

We answer with two coordinates. For scale s and HR frequency r (cycles per HR pixel), let $u = 2sr$ be the frequency relative to the LR Nyquist limit. Our claims are:

1. **Recovery lives in u.** Per-band recovery $R(u)$ is nearly scale-independent and decoder-independent. Training, encoder capacity and decoder capacity change it only above $u \approx 0.75$; decoder design does not.

2. **Conditioning acts in r.** Changing the cell multiplies the native output by an approximately scale-independent low-pass $w(r)$. Its cost is bounded by the recovered energy at those r, so the cell's leverage decays with scale.

3. **Consequence.** The cell input can be dropped at large scales, and decoder design is not a lever. The capacity that learns the above-Nyquist prior, whether in the encoder or in the decoder MLP, is. We test this with matched training.

That cell inputs can occasionally hurt is already known [3]. Our contribution is a quantitative account of where any decoder- or cell-side change can act at all. Every experiment followed a frozen protocol (code, predictions and scoring hashed before results); Table 2 lists outcomes, including failures.

2. SETUP

Models. LIIF, LTE and SC-INR (an LTE-derived variant whose cell enters through an analytic sinc weight) share an EDSR-baseline encoder [13] and the LIIF training recipe (DIV2K [14], scales $\mathcal{U}(1, 4)$, 1000 epochs, three seeds). We add GaussianSR [6] (EDSR-baseline encoder) and RDN-BTC [15, 16]. For matched comparisons we train LIIF for 200 epochs with an identical recipe and EDSR-baseline, RDN or SwinIR [17] encoders, a reduced decoder (2×64 MLP) or no cell input.

Data and degradation. Fresh DIV2K valid 0869–0900 (never used in this project), Set14 [18], BSD100 [19] and Urban100 [20]. We take centre 240^2 crops and bicubic anti-aliased downsampling at $\times\{2, 3, 4, 6\}$; $\times 6$ lies outside training support.

Measurements. After a 4-pixel shave, errors are split by Chebyshev FFT radius into u-bins of width 0.25. Then $R(u) = 1 - \mathrm{MSE}_u/E_u$, with E_u the target energy in the bin. The cell sweep multiplies the native cell by $m \in [\frac{1}{2}, 2]$ (11 values).

3. RECOVERY LIVES IN U

Fig. 1a shows $R(u)$ for four scales. Below the LR Nyquist limit ($u < 1$) networks and bicubic interpolation both recover most energy. Between $u = 0.75$ and 2 the network recovers far more: at $u = 1$, 0.68 versus 0.20 for bicubic; at $u = 1.5$, 0.36 versus 0.04. Curves at different scales nearly coincide, but only approximately. Bicubic interpolation collapses in u as well, so the collapse itself mainly reflects approximately scale-invariant image statistics and a degradation defined in LR coordinates. What matters below is which parts of the network can move the curve. Our frozen tolerance (spread ≤ 0.05) held in 45/54 crop cases and in 53/72 full-image cases; the misses concentrate near the HR Nyquist limit at $\times 2$.

Decoders do not move $R(u)$. On a shared EDSR-baseline encoder, LIIF, LTE, SC-INR and GaussianSR differ by at most 0.010 in every bin (18/18 and 18/18; Fig. 1b). Correspondingly, their PSNR differs by only 0.03–0.06 dB, shrinking from $\times 2$ to $\times 6$. On full Set14, BSD100 and Urban100 images the four decoders stay within 0.019 in every bin $u \in [0.5, 2)$ (54/54), and all of them exceed bicubic by 0.31–0.59 at $u \in [1, 1.25)$.

Training and capacity do. Between epoch 100 and 1000, $R(u)$ rises by up to 0.045 for $u \geq 1$ (12/12 bins). It is unchanged below $u < 0.75$ (≤ 0.008; 9/9; Fig. 1c). With decoder and recipe fixed, an RDN encoder raises $R(u)$ by up to 0.053 for $u \geq 1$ (12/12) and leaves $u < 0.75$ unchanged (9/9). A SwinIR encoder does the same at an equal 50 epochs (12/12, 9/9; Fig. 1c). External checkpoints (RDN-BTC, official EDSR) underperformed under our degradation and are not used as encoder evidence.

4. CONDITIONING ACTS IN R

Let P_1 be the native output and P_m the output with cell multiplied by m. Projected per band, we find

$$P_{m,b} \approx w_m(r_b)\, P_{1,b}, \quad w_m(r) \in \mathbb{R}, \tag{1}$$

a low-pass in HR frequency whose shape depends only weakly on scale (Fig. 2a). Writing $e_b = T_b - P_{1,b}$ for the native error, Eq. (1) gives the exact expansion

$$\Delta\mathrm{MSE}_b = 2(1-w)\,\mathrm{Re}\langle e_b, P_{1,b}\rangle + (1-w)^2\|P_{1,b}\|^2. \tag{2}$$

If the native output were the conditional mean, the orthogonality principle would give $\langle e_b, P_{1,b}\rangle \approx 0$. Two things follow. The first-order term vanishes, which is what calibration of the native cell means. The emitted energy equals the recovered energy, $\|P_{1,b}\|^2 \approx R_b E_b$. Hence

$$\Delta\mathrm{MSE}(s) \approx \sum_r \big(1 - w(r)\big)^2\, R(2sr)\, E(r). \tag{3}$$

The two coordinates meet in Eq. (3): the cell supplies a filter in r, the network supplies recovery in $u = 2sr$. Because R decreases in u, the same filter acts on ever less recovered energy as s grows.

Checking the assumptions. The models are trained with an L_1 loss, so orthogonality can only hold approximately. A frozen replication on the fresh DIV2K images with all three LIIF-family decoders supports the approximation. $|\langle e_b, P_{1,b}\rangle|/\|P_{1,b}\|^2 \leq 0.15$ in all 105 bins with $u < 2$. The minimum of $w_2(r)$ lies at $r = 0.28$–0.34 for every decoder at $\times 2$–$\times 4$, drifting slightly lower as the scale grows (9/9). For $m = 2$, Eq. (3) predicts the measured cost at $\times 3$ and $\times 4$ to within a factor 0.71–0.84 (6/6). It underpredicts consistently, because the cell also changes the output beyond a per-band scalar; on Set14 the shortfall reaches about $2\times$ at $\times 2$.

Tests. For $m = 2$, the cost peaks at $r \approx 0.25$ for every model, seed and scale (36/36; Fig. 2b). Its magnitude tracks recovered energy (Spearman 0.77–0.78). The second-order term ranks per-band cost with $\rho = 0.86$–0.91 and accounts for 70–77% of it. Cost decays monotonically with scale for all nine model–seed pairs (Fig. 2c), and also for GaussianSR and RDN-BTC. The native cell is close to calibrated: the best fixed multiplier improves on it by at most 0.017 dB. Its optimum drifts slightly above $m = 1$ at $\times 4$–$\times 6$, so the frozen "native is optimal" gate failed (20/27).

Limits of the model. For small mismatch ($m = 4/3$) the first-order term dominates and Eq. (1) does not predict cost. The peak at $r \approx 0.25$ holds only for the LIIF family; GaussianSR and BTC peak elsewhere. We had also predicted that a too-small cell (requesting detail) would cost more than a too-large one, by analogy with prior-mismatch asymmetry. The opposite held (33/144), so we withdraw that hypothesis.

5. CONSEQUENCE: WHAT IS A LEVER

Before training we froze three design predictions (Table 1). (D2) Removing the cell input costs little at large scales. With two seeds it costs 0.10 dB at $\times 2$ and is indistinguishable from zero from $\times 3$ on. The seed-1 gate still failed narrowly, because $\times 4$ exceeded $\times 3$ and the 0.03 dB bound. (D3) A stronger encoder helps more than a reduced decoder hurts. This held: RDN adds 0.23 dB at $\times 4$, and both RDN and SwinIR raise $R(u)$ only above the LR Nyquist limit. (D1) A $4\times$ smaller decoder MLP costs ≤ 0.05 dB at $\times 4$/$\times 6$. This *failed*: it costs 0.11–0.21 dB at every scale (both seeds), again only for $u \geq 0.75$. The decoder MLP therefore also stores part of the above-Nyquist prior. The correct reading is that decoder *design* is not a lever (four designs, identical $R(u)$), while learned capacity

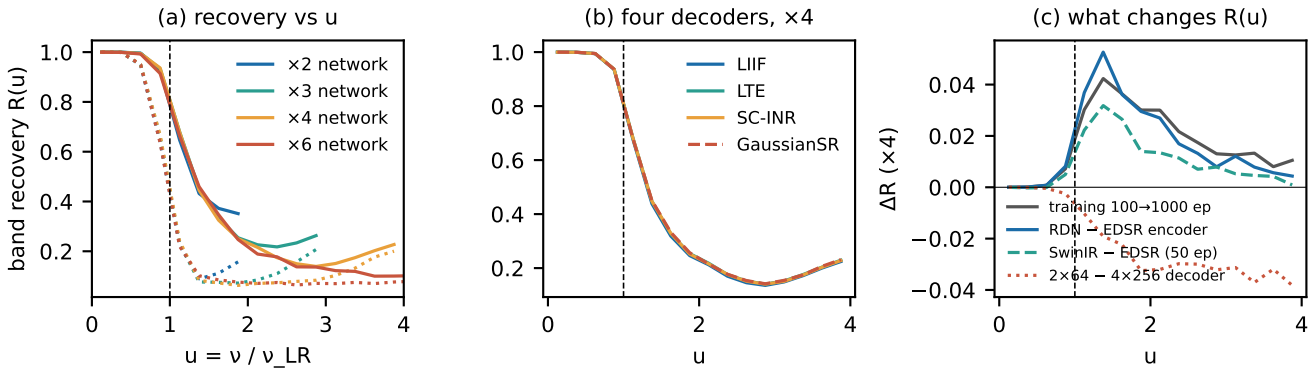


Fig. 1. (a) Band recovery $R(u)$ at four scales (solid: mean of three decoders and three seeds; dotted: bicubic). (b) Four decoders on the same encoder. (c) Change in $R(u)$ at $\times 4$ from longer training, stronger encoders and a smaller decoder MLP.

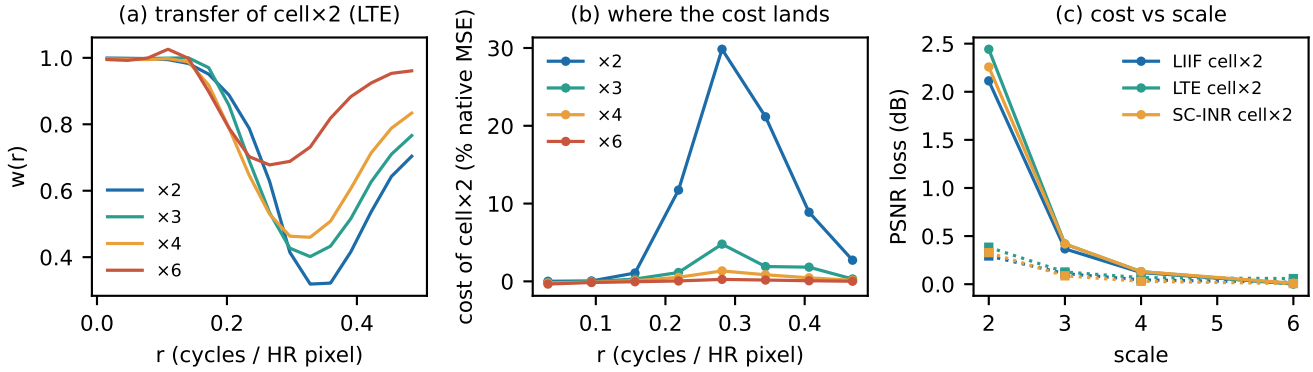


Fig. 2. (a) Spectral transfer $w(r)$ of doubling the cell (LTE, Set14). (b) Where its cost lands in HR frequency. (c) PSNR cost of doubling (solid) or halving (dotted) the cell, by scale.

Table 1. Matched LIIF training (200 epochs, mean of two seeds): PSNR change (dB, non-DC) relative to EDSR-baseline LIIF on DIV2K 0869–0900. All 95% image-bootstrap intervals exclude 0 except °. †One seed, compared with EDSR at an equal 50 epochs.

Arm	×2	×3	×4	×6
decoder 2×64	−0.21	−0.17	−0.13	−0.11
no cell input	−0.10	−0.03°	−0.02°	−0.00°
RDN encoder	+0.30	+0.23	+0.23	+0.18
SwinIR encoder†	+0.17	+0.18	+0.12	+0.07

anywhere in the network is. Per parameter, the decoder MLP is the more efficient place for it (about 0.3M parameters for 0.15 dB, against about 20M for RDN's 0.23 dB). The scale-dependent cell is the part that can be dropped at large scales.

6. LIMITATIONS AND CONCLUSION

We study bicubic degradation, centre crops (full images for the decoder test), convolutional and Transformer encoders and LIIF-style training. The u-collapse is approximate. The filter model of the cell holds for large mismatch only. Its frequency locus is specific to decoder families. Within these limits, ASISR gains lie in what the network learns above the LR Nyquist limit, and learned capacity, not decoder form, sets that prior. The cell is an HR-frequency control whose reach is the recovered energy at the frequencies it filters. Eq. (3) therefore tells in advance at which scales a new cell or footprint design can matter: only where $R(2sr)$ is still substantial at the frequencies the design changes.

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Table 2. Pre-registered predictions and outcomes.

Prediction	Outcome
$R(u)$ spread across scales ≤ 0.05	45/54, 53/72 **fail**
decoders within 0.03 (four decoders)	18/18, 18/18
same, full benchmark images	54/54
training raises only $u \geq 1$	12/12; 9/9
cell$\times 2$ cost peaks at $r \in [0.19, 0.375)$	36/36 (LIIF family)
cost $\sim$ recovered energy ($\rho \geq 0.7$)	0.77–0.78
leverage decays with scale	9/9
law Eq. (3) within $\times[0.7, 1.3]$ ($\times 3/\times 4$)	6/6
$\lvert\langle e, P_1\rangle\rvert / \lVert P_1\rVert^2 \leq 0.15$	105/105
native cell optimal	20/27 **fail**
too-small cell costs more	33/144 **fail**
$\geq$80% of cost at $u \geq 1$	75% **fail**
RDN raises only $u \geq 1$ (T1–T3)	12/12; 9/9; pass
SwinIR raises only $u \geq 1$ (ep. 50)	12/12; 9/9; pass
small decoder costs $\leq$ 0.05 dB at $\times 4$+ (D1)	**fail**
no-cell cost decays, $\leq$ 0.03 dB at $\times 4$+ (D2)	**fail** (narrow)
encoder gain $>$ decoder loss (D3)	pass
seed-2 replication of all arms (K1–K3)	11/12; 12/12, 9/9; 12/12, 9/9

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